

# Supporting Information: Auditing Whether Models Learn RNA Sequence or Database Structure in RNALocate

Yicheng "Billy" Lu<sup>1</sup>; Xinyi Lu<sup>2</sup>

## Supplementary Methods

### Frozen analysis plan

The context-shortcut audit was frozen on 14 August 2026 before the new analyses were run. Existing upstream annotation-recovery results were already known, so the plan is a supplemental post hoc analysis plan rather than a preregistration.

### Bootstrap

Average-precision intervals and contrasts used 1,000 paired resamples of all-sequence graph components. Component resampling preserves within-component dependence and repeated contexts belonging to a gene.

### Support-gradient intervals

Observed fractions at support values 0-15 used Wilson 95% binomial intervals. Support zero is a cohort-selection boundary and is not interpreted as the first point of a causal dose-response curve.

### Permutation control

Each of 200 replicates permuted support among genes within each context and split, repeated hyperparameter selection, refit on training plus validation, and evaluated held-out components.

### Calibration

Context, context-plus-support, and full-model predictions were separated into ten equal-frequency test bins. Predicted probability was compared with observed fraction and Wilson intervals.

### Sequence-view sensitivity

Composition plus log length, normalized 3-mer frequencies, and the complete 70-feature representation were evaluated under the same partition, tuning grid, and logistic-regression family.

## Supplementary Tables

Table S1 | Context-specific context-plus-support performance

| Context  | Pairs | Prev.  | AP     | AP lift | AUROC  | Brier  |
|----------|-------|--------|--------|---------|--------|--------|
| A549     | 2822  | 0.7101 | 0.9420 | 0.2318  | 0.9041 | 0.1011 |
| GM12878  | 2822  | 0.7381 | 0.9462 | 0.2081  | 0.8871 | 0.1160 |
| H1-hESC  | 2822  | 0.8795 | 0.9901 | 0.1105  | 0.9409 | 0.0612 |
| HEK293T  | 2822  | 0.9745 | 0.9922 | 0.0177  | 0.8067 | 0.0258 |
| HT1080   | 2822  | 0.6687 | 0.9662 | 0.2975  | 0.9464 | 0.0877 |
| HUVEC    | 2822  | 0.7158 | 0.9783 | 0.2625  | 0.9561 | 0.0743 |
| HeLa-S3  | 2822  | 0.6421 | 0.9205 | 0.2784  | 0.8983 | 0.1180 |
| HepG2    | 2822  | 0.7155 | 0.9655 | 0.2501  | 0.9275 | 0.0986 |
| IMR-90   | 2822  | 0.3629 | 0.4768 | 0.1139  | 0.6740 | 0.2290 |
| K562     | 2822  | 0.6545 | 0.9226 | 0.2681  | 0.8899 | 0.1258 |
| MCF-7    | 2822  | 0.7555 | 0.9752 | 0.2198  | 0.9350 | 0.0881 |
| NCI-H460 | 2822  | 0.5595 | 0.8751 | 0.3155  | 0.8977 | 0.1165 |
| NHEK     | 2822  | 0.7300 | 0.9730 | 0.2430  | 0.9381 | 0.0885 |
| SK-MEL-5 | 2822  | 0.5829 | 0.8974 | 0.3145  | 0.9028 | 0.1168 |
| SK-N-DZ  | 2822  | 0.6244 | 0.8959 | 0.2716  | 0.8819 | 0.1265 |
| SK-N-SH  | 2822  | 0.6882 | 0.9201 | 0.2320  | 0.8756 | 0.1232 |

Table S2 | Low-level sequence representation sensitivity

| Feature view       | Features | Model         | C       | AP     | AUROC  | Brier  |
|--------------------|----------|---------------|---------|--------|--------|--------|
| composition_length | 6        | sequence_only | 10.0000 | 0.7870 | 0.6689 | 0.1963 |
| composition_length | 6        | full          | 0.0100  | 0.9577 | 0.9156 | 0.1059 |
| three_mer_only     | 64       | sequence_only | 1.0000  | 0.8139 | 0.7034 | 0.1890 |
| three_mer_only     | 64       | full          | 0.0100  | 0.9580 | 0.9158 | 0.1059 |
| complete_70        | 70       | sequence_only | 10.0000 | 0.8234 | 0.7153 | 0.1854 |
| complete_70        | 70       | full          | 0.0100  | 0.9580 | 0.9158 | 0.1058 |

Table S3 | Incremental AP effects

| Contrast                       | AP difference | CI low | CI high | Replicates |
|--------------------------------|---------------|--------|---------|------------|
| sequence_after_context         | 0.0787        | 0.0719 | 0.0863  | 1000       |
| support_after_context          | 0.1535        | 0.1458 | 0.1616  | 1000       |
| sequence_after_context_support | 0.0009        | 0.0007 | 0.0010  | 1000       |
| support_after_context_sequence | 0.0757        | 0.0695 | 0.0823  | 1000       |

**Table S4 | Split sensitivity across annotation-recovery and record-availability analyses**

| Partition       | Method                           | Comp. | Largest | Annotation:<br>Context AP | Annotation:<br>3-mer AP | Availability:<br>C+S AP | Availability:<br>Full AP |
|-----------------|----------------------------------|-------|---------|---------------------------|-------------------------|-------------------------|--------------------------|
| primary_g80_c80 | exhaustive threshold graph       | 18437 | 14      | 0.7198                    | 0.3054                  | 0.9572                  | 0.9580                   |
| cluster_g70_c80 | MMseqs2 easy-cluster stress test | 18120 | 21      | 0.7205                    | 0.3055                  | 0.9576                  | 0.9584                   |
| cluster_g60_c80 | MMseqs2 easy-cluster stress test | 17887 | 21      | 0.7201                    | 0.3052                  | 0.9570                  | 0.9579                   |
| cluster_g80_c50 | MMseqs2 easy-cluster stress test | 18019 | 15      | 0.7197                    | 0.3054                  | 0.9576                  | 0.9585                   |

**Table S5 | Support-zero exclusion sensitivity**

| Model                        | Pairs | Prev.  | AP     | CI low | CI high | AUROC  | Brier  |
|------------------------------|-------|--------|--------|--------|---------|--------|--------|
| Context                      | 45003 | 0.6866 | 0.8014 | 0.7923 | 0.8115  | 0.6560 | 0.1987 |
| Other-context support        | 45003 | 0.6866 | 0.9006 | 0.8959 | 0.9051  | 0.8558 | 0.1302 |
| Context + support            | 45003 | 0.6866 | 0.9573 | 0.9544 | 0.9601  | 0.9159 | 0.1059 |
| Context + support + sequence | 45003 | 0.6866 | 0.9582 | 0.9552 | 0.9610  | 0.9164 | 0.1057 |